each crossing or tangent point. For **a** and **b**, the least time gap is the start delay for a meeting. Trace the paths in Fig. 20-4-1 for **a** and **b**.

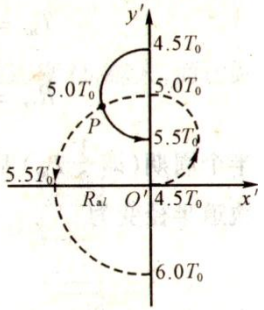


Fig. 20-4-2

A crossing between $5T_0$ and $5.25T_0$ (point P, Fig. 20-4-1) gives the least gap. For **a**, **b**, before $5T_0$ or after $5.25T_0$, other points give **a** and **b** longer time gaps.

On the negative y-axis, $3R_{al}$ from O ($6R_{bl}$), take O' as origin (Fig. 20-4-2). Let $t = 5T_0$ be time zero: $t' = 0$ for **a**, $t'' = 0$ for **b**. Left of y, **a**, **b** share a period; ω_1 is **a**, **b**'s common angular speed left of y:

$$\omega_1 = \frac{\pi}{T_0} \tag{12}$$

For **a**:

$$x_a' = R_{al}\cos\left(\omega_1 t' + \frac{\pi}{2}\right) \tag{13}$$

$$y_a' = R_{al}\sin\left(\omega_1 t' + \frac{\pi}{2}\right) \tag{14}$$

For **b**:

$$x_b' = \frac{R_{al}}{2}\cos(\omega_1 t'' + \pi) \tag{15}$$

$$y_b' = R_{al} + \frac{R_{al}}{2}\sin(\omega_1 t'' + \pi) \tag{16}$$

At P:

$$x_a' = x_b' \qquad y_a' = y_b' \tag{17}$$

Solving gives the required time difference:

$$\Delta t = t' - t'' = 0.080\,2\frac{\pi m}{qB_0} \tag{18}$$

If **a**, **b** start together, **a** follows **b** by $0.080\,2\,\pi m/qB_0$ at P. To make **a**, **b** meet, **b** must start later than **a** by $0.080\,2\,\pi m/qB_0$.

These equations also give coordinates of O:

$$x_P' = -\frac{\sqrt{15}}{8}R_{al} = -0.484R_{al} = -0.968\frac{mv_0}{qB_0} \tag{19}$$

$$y_P' = -\frac{7}{8}R_{al} = 0.875R_{al} = 1.75\frac{mv_0}{qB_0} \tag{20}$$

In the Oxy coordinate system:

$$x_P = -0.968\frac{mv_0}{qB_0} \tag{21}$$

$$y_P = -3R_{al} + \frac{7}{8}R_{al} = -2.125R_{al} = -4.25\frac{mv_0}{qB_0} \tag{22}$$

5.

1. Fig. 20-5-1 shows a myopic eye viewing its farthest clear object. It can accommodate to see up to a,; axial points form images on the retina.

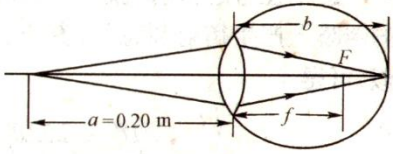


Fig. 20-5-1

An axial point at infinity focuses only at F, leaving a blurred patch on the retina. A card with a small circular hole near the eye reduces the patch (Fig. 20-5-2). A smaller patch gives a clearer image. Thus, near the eye, a hole